Electronics and Electrical Communications Engineering Department

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Neural Networks

Assignment 1
Classical Machine Learning Methods

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1 Perceptron Learning Rule Derivation

In this section, we derive the perceptron learning rule step by step starting from the model definition.

1.1 Perceptron Model

The perceptron computes a linear combination of the input features:

$$a = \mathbf{w}^T\mathbf{x} + b$$

The predicted label is given by:

$$\hat{y} = \text{sign}(a)$$

Where:

- $\mathbf{x}$: Input feature vector
- $\mathbf{w}$: Weight vector
- b: Bias term
- a: Activation (linear combination)
- $\hat{y}$: Predicted label
- y: True label ($y \in \{+1, -1\}$)

1.2 Correct vs Incorrect Prediction

A prediction is considered correct if:

$$y(\mathbf{w}^T\mathbf{x} + b) > 0$$

and incorrect if:

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0$$

Explanation:

- If $y = +1$, then we want $(\mathbf{w}^T\mathbf{x} + b) > 0$
- If $y = -1$, then we want $(\mathbf{w}^T\mathbf{x} + b) < 0$

Thus, multiplying by y ensures that a correct classification results in a positive value.

1.3 Objective

If the prediction is incorrect, we need to:

- Update the weights $\mathbf{w}$
- Update the bias b
- Move the decision boundary toward the correct classification

1.4 Proposed Update Rule

We propose the following update rule:

$$\mathbf{w}_{new} = \mathbf{w} + y\mathbf{x}$$

$$b_{new} = b + y$$

Condition: Apply the update only if:

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0$$

1.5 Why Does This Work?

We analyze how the update affects the classification score:

$$y(\mathbf{w}_{new}^T\mathbf{x} + b_{new})$$

Substitute the updated values:

$$= y\left((\mathbf{w} + y\mathbf{x})^T\mathbf{x} + (b + y)\right)$$

Expand:

$$= y\left(\mathbf{w}^T\mathbf{x} + y\mathbf{x}^T\mathbf{x} + b + y\right)$$

$$= y(\mathbf{w}^T\mathbf{x} + b) + y^2(\mathbf{x}^T\mathbf{x} + 1)$$

Since:

$$y^2 = 1$$

we get:

$$= y(\mathbf{w}^T\mathbf{x} + b) + (\|\mathbf{x}\|^2 + 1)$$

Conclusion:

- The term $(\|\mathbf{x}\|^2 + 1)$ is always positive
- Therefore, the value increases after the update
- This moves the prediction toward the correct classification

1.6 Final Learning Rule

Weight update:

$$\mathbf{w} \leftarrow \mathbf{w} + y\mathbf{x}$$

Bias update:

$$b \leftarrow b + y$$

Apply update only when:

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0$$

1.7 Interpretation

The perceptron learning rule:

- Adjusts the decision boundary only when a mistake occurs
- Moves the boundary toward correctly classifying the input
- Reinforces correct classification over iterations

This simple yet effective rule forms the foundation of many modern machine learning algorithms.

2 Introduction

This report presents a semi-supervised pipeline for labeling a dataset of 10,000 handwritten digit images. The objective is to achieve high classification accuracy while minimizing manual labeling effort by combining clustering, feature extraction, and active learning.

3 Pipeline Overview

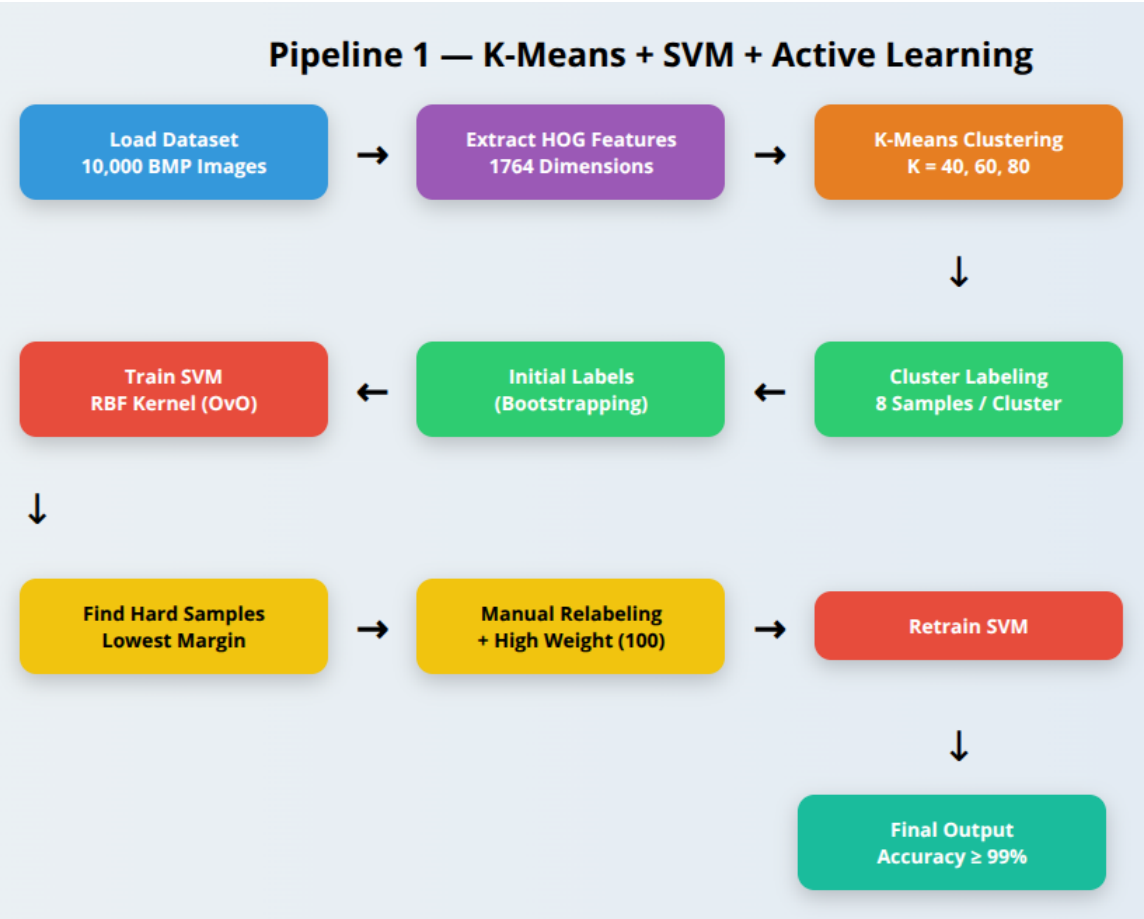

Figure 1: Pipeline 1: K-Means + SVM + Active Learning

The pipeline consists of the following steps:

- Load dataset (10,000 BMP images)
- Extract HOG features
- Apply K-Means clustering
- Label clusters (bootstrapping)
- Train SVM classifier
- Apply active learning for refinement

4 Feature Extraction

Histogram of Oriented Gradients (HOG) was used to extract features from images.

- Image size: 32×32
- Feature vector size: 1764

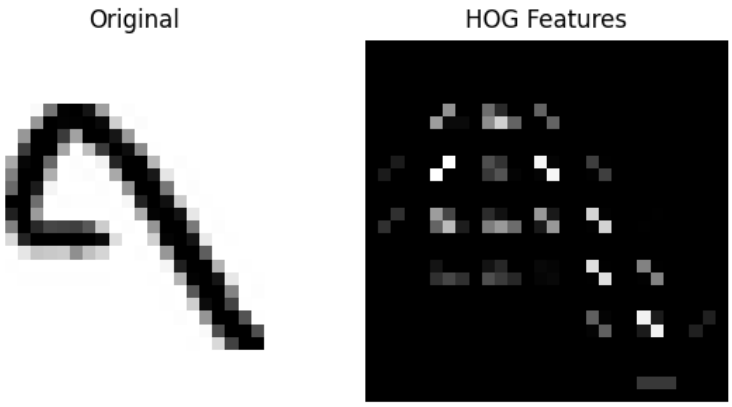

Figure 2: Example of HOG Feature Representation

5 K-Means Clustering

K-Means clustering was applied with different values of K:

- K = 40
- K = 60
- K = 80

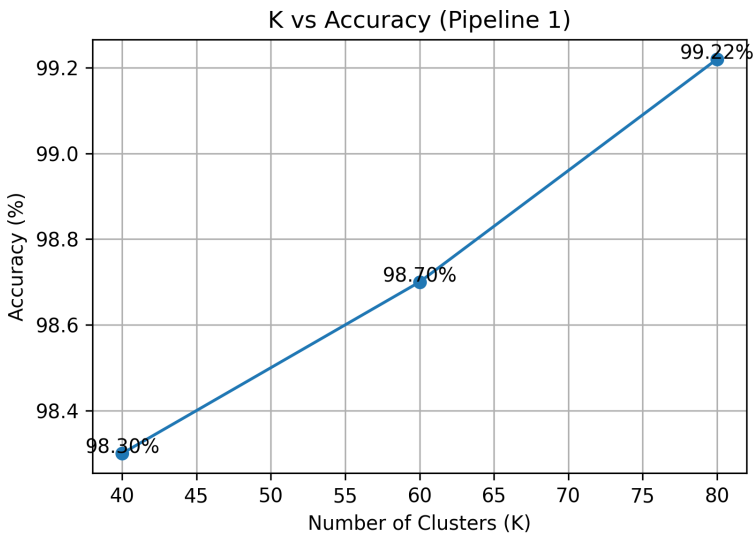

Figure 3: K vs Accuracy Before SVM

6 SVM and Active Learning

A multi-class SVM with RBF kernel (One-vs-One strategy) was used.
Active learning was applied as follows:

- Select 30 least confident samples (lowest margin)
- Relabel manually
- Assign weight = 100
- Retrain SVM

The process stops when:

- Accuracy $\geq 99\%$
- Improvement $< 0.1\%$

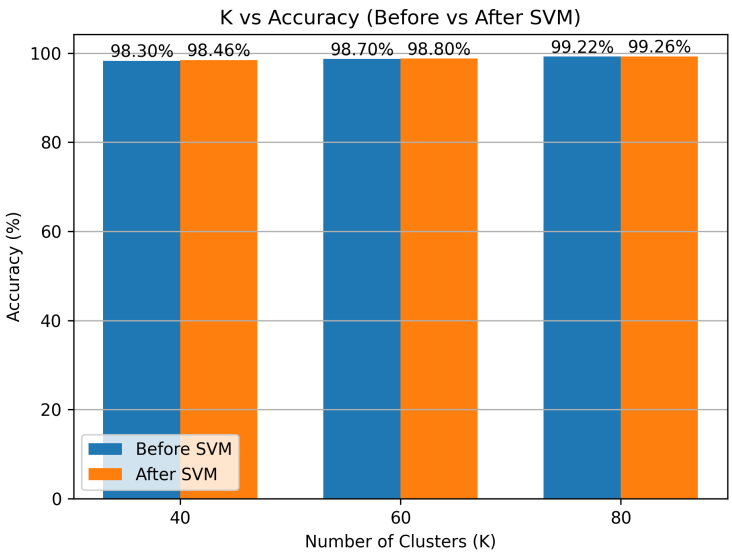

Figure 4: Accuracy Before vs After SVM

7 Results

7.1 Accuracy

K	Before SVM (%)	After SVM (%)
40	98.30	98.46
60	98.70	98.80
80	99.22	99.26

Table 1: Accuracy Comparison

7.2 Iterations

K	Iterations
40	2
60	2
80	2

Table 2: Number of Iterations

8 Manual Effort

For K = 80:

- Cluster labeling: $80 \times 8 = 640$ images
- Active learning: $30 \times 2 = 60$ images

Total images manually handled:

$$700 \text{ images}$$

Total time:

- Cluster labeling: 1600 seconds
- Active learning: 600 seconds

$$\text{Total time} = 2200 \text{ seconds} \approx 36.7 \text{ minutes}$$

9 Conclusion

The proposed pipeline significantly reduces manual labeling effort from 27.8 hours to approximately 37 minutes while achieving 99.26% accuracy.

K-Means clustering provides strong initial labels, while SVM and active learning refine ambiguous samples. The improvement from SVM is more noticeable at lower K values, while higher K values already produce highly pure clusters.

10 Pipeline 2: Manual Seed + Augmentation + Self-Training

This pipeline adopts a semi-supervised learning approach starting from a small, high-quality human-labelled seed set. The model progressively expands its knowledge using data augmentation, active learning, and self-training.

10.1 Pipeline Diagram

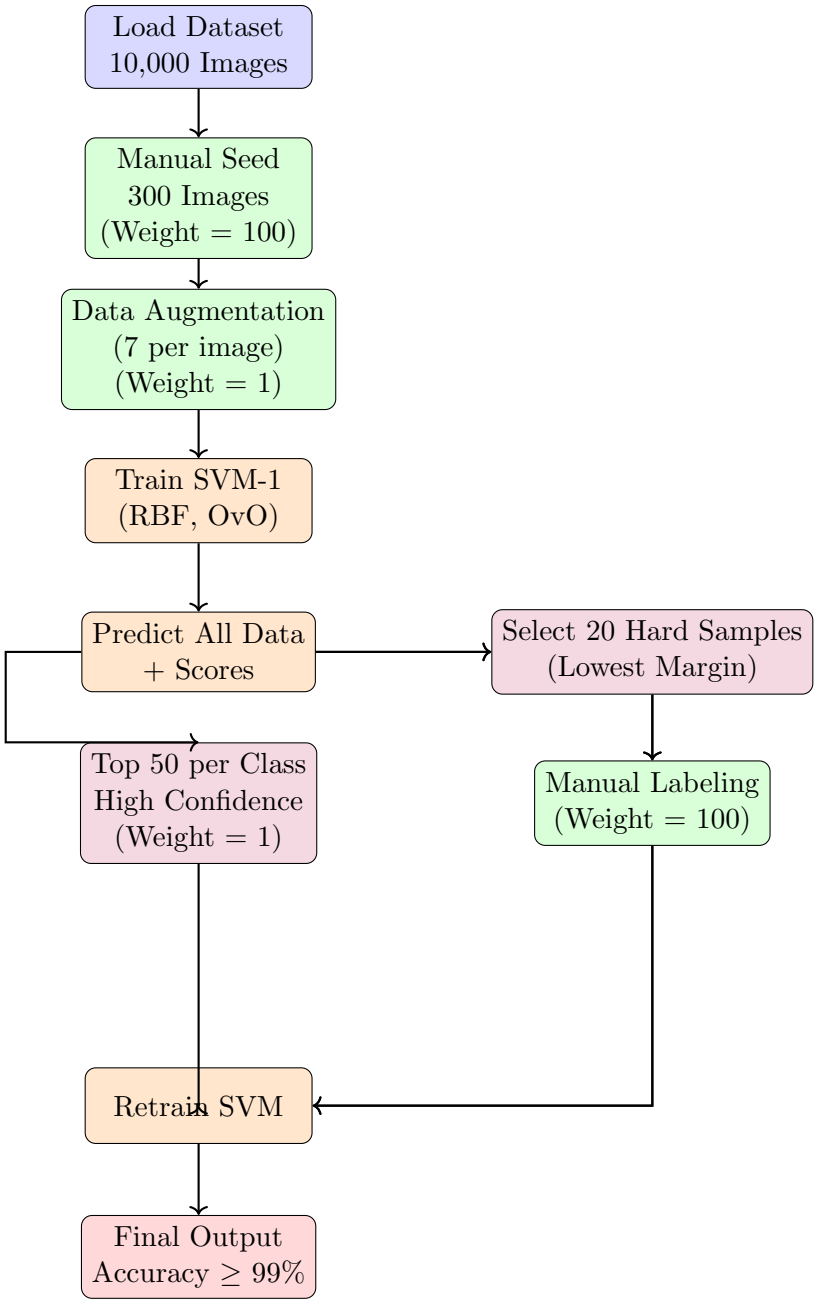

Figure 5: Pipeline 2: Semi-Supervised Learning Framework

10.2 Final Results

The final performance of Pipeline 2 after convergence is summarized as follows:

- Final Accuracy: 99.04%
- Number of Iterations: 4
- Total Manual Labels: 380
- Total Pseudo Labels: 1898

- Estimated Manual Time: 63.33 minutes

10.3 Iteration Progress

The improvement in model accuracy across iterations is shown in Figure below.

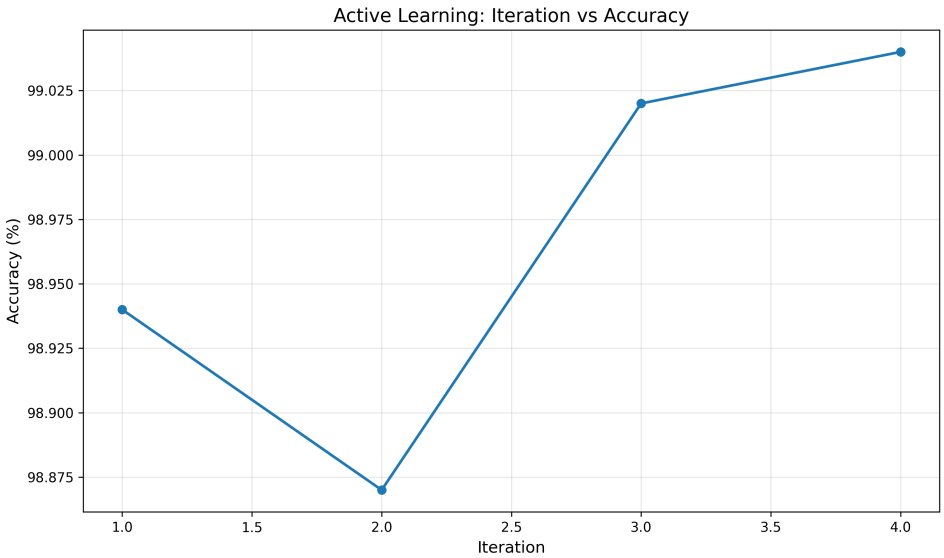

Figure 6: Active Learning: Iteration vs Accuracy

10.4 Visualization

Figure 7: Initial Human Seed Samples (10 per class)

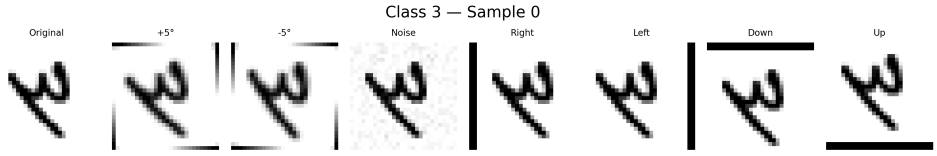

Figure 8: Example of Data Augmentation (7 variants)

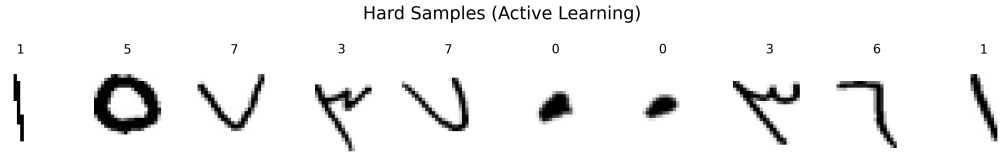

Figure 9: Hard Samples (Iteration 1)

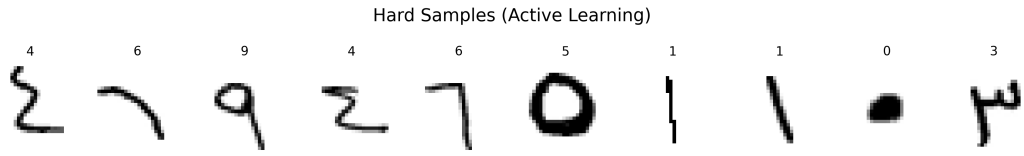

Figure 10: Hard Samples (Iteration 2)

10.5 Manual Effort

Manual labeling consists of:

- Initial seed: 300 images
- Active learning: $20 \times 4 = 80$ images

Total manual labels:

$$300 + 80 = 380$$

Estimated time:

$$380 \times 10 = 3800 \text{ seconds} \approx 63.33 \text{ minutes}$$

10.6 Comparison with Pipeline 1

Metric	Pipeline 1	Pipeline 2
Accuracy	99.26%	99.04%
Manual Images	700	380
Manual Time	36.7 min	63.3 min
Iterations	2	4

Table 3: Pipeline Comparison

Pipeline 1 achieves higher accuracy due to clustering-based initialization, while Pipeline 2 significantly reduces manual labeling effort by leveraging self-training. However, Pipeline 2 requires slightly more total time due to per-image labeling cost.

10.7 Discussion

Pipeline 2 demonstrates the effectiveness of semi-supervised learning. Starting from a small, high-quality seed set, the model successfully expands its training data using both active learning and self-training.

Although the final accuracy (99.04%) is slightly below Pipeline 1, the reduction in manual labeling effort is significant. This pipeline better reflects real-world scenarios where labeled data is scarce and expensive.

The combination of:

- Human supervision (high confidence)
- Model confidence (self-training)
- Iterative refinement

creates a scalable and efficient labeling strategy.

11 References

- Project Repository:
  `https://github.com/youefkh05/Neural_Netwok_projects`